# Reports and Outputs of Scan Insertion

# REPORTS GENERATED DURING SCAN INSERTION

- Scan chain report
- Scan cell report

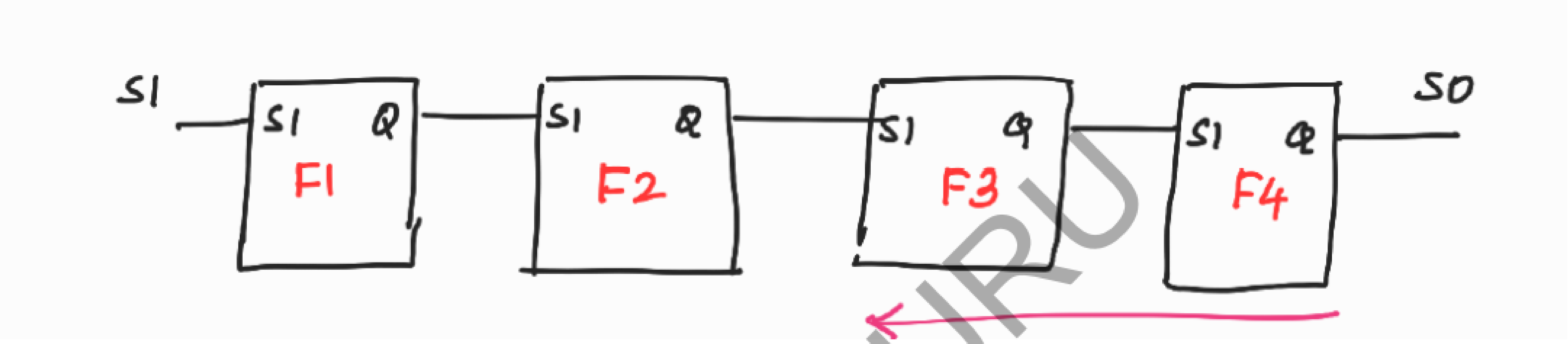
# Scan Chain Report

- no. of scan chains

Has the following information for all the scan chains

- Scan input, Scan output, no. of flops (length of scan chain), scan enable, set, reset, clock

# Scan Cell Report






**Actual Order :**

1st flop -> F1

2nd flop -> F2

3rd flop -> F3

4th flop -> F4

**Scan Cell Report :**

1st flop -> F4

2nd flop -> F3

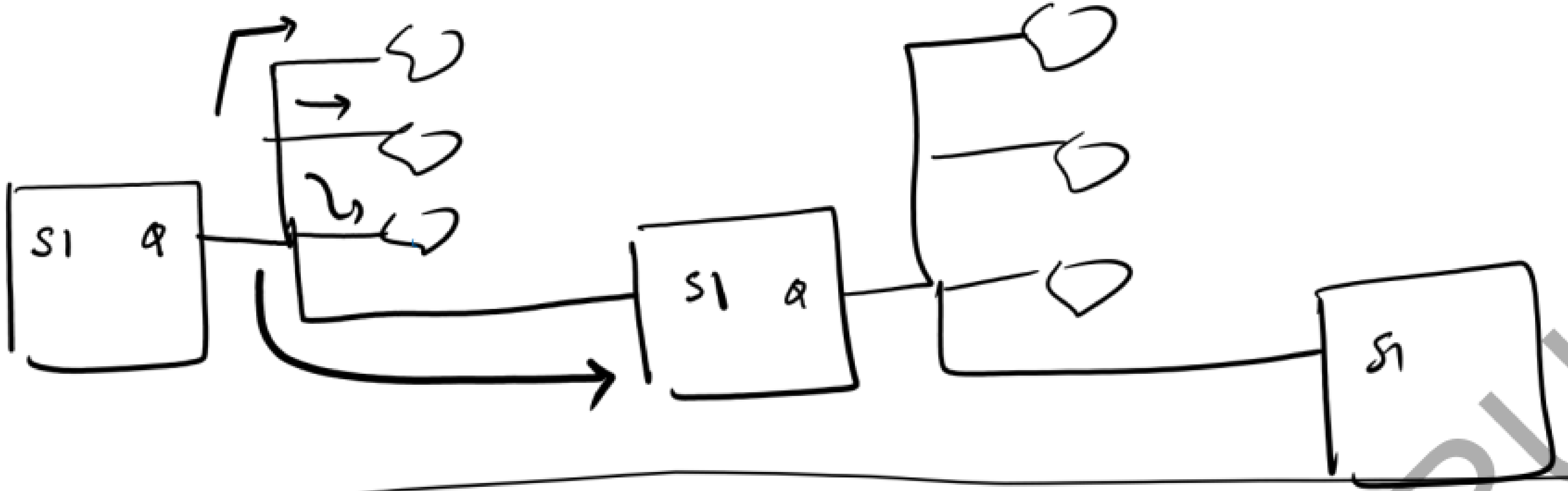
3rd flop -> F2

4th flop -> F1

from i/p to o/p → it is difficult

∵ Q of a flop has many fanouts.

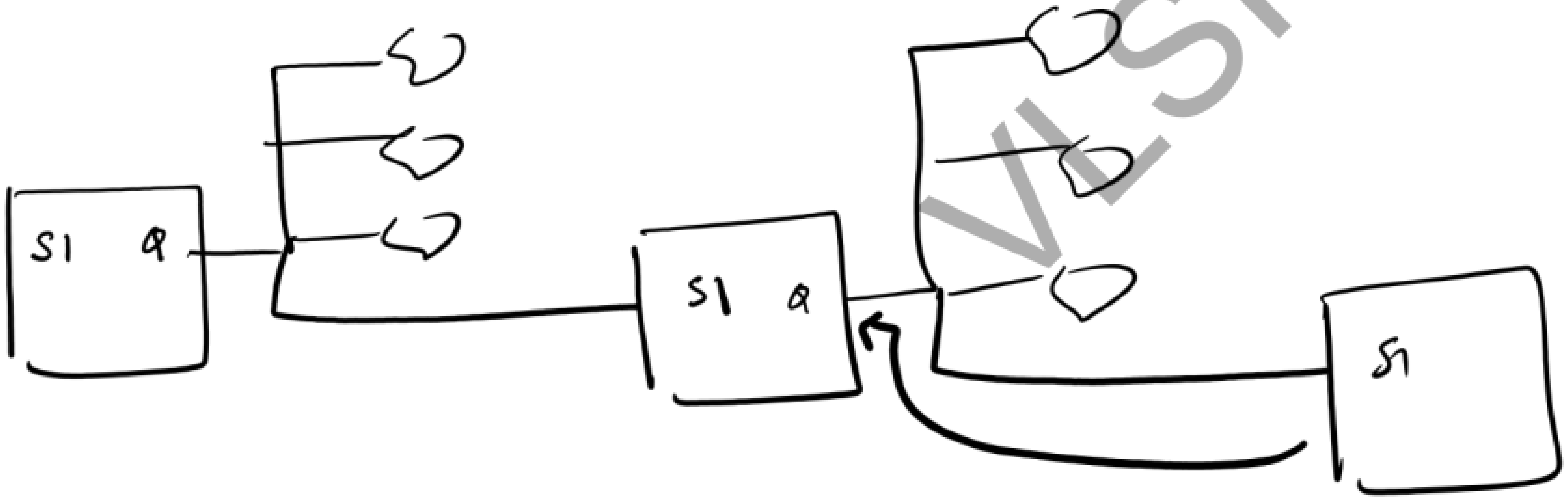
∴ One has to ~~...~~ trace multiple fanouts for reaching the SI of the next flop.




from o/p to I/p ↳ easier.

∵ SI of a flop has only 1 fanin (o/p of the previous flop)

∴ One has to trace only 1 connection



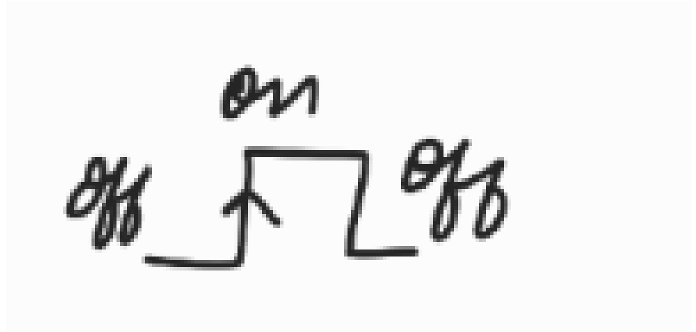
last ff of chain1

| CellNo | Chain Name | Group Name | Pathname | ShiftReg ID/CellNo | Library ModelName | ScanOut | Clock Pinname | Clock Polarity |
|---|---|---|---|---|---|---|---|---|
| 0 | chain1 | dummy | /CountWr_r_reg_1_ | -/- | SDFFRHQX4 | Q | FastClk | (+) |
| 1 | chain1 | dummy | /StartWrClkd_reg | 1/2 | DFFRX1 | Q | FastClk | (+) |
| 2 | chain1 | dummy | /StartWr_reg | 1/1 | SDFFRX1 | Q | FastClk | (+) |
| 3 | chain1 | dummy | /StartSyncWr_reg | 2/2 | DFFRX1 | QN | FastClk | (+) |
| 4 | chain1 | dummy | /MemWrSync1_reg | 2/1 | SDFFRX1 | Q | FastClk | (+) |
| 5 | chain1 | dummy | /DMSel_r_reg | -/- | SDFFSX4 | QN | FastClk | (+) |
| 6 | chain1 | dummy | /CurrentState_reg_2_ | -/- | SDFFRX4 | QN | FastClk | (+) |
| 7 | chain1 | dummy | /CurrentState_reg_1_ | -/- | SDFFRX4 | QN | FastClk | (+) |
| 8 | chain1 | dummy | /PMSel_r_reg | -/- | SDFFSX1 | QN | FastClk | (+) |
| 9 | chain1 | dummy | /PMWr_r_reg | -/- | SDFFSX1 | QN | FastClk | (+) |

1st ff of chain1

# Scan Cell Report (Contd…)

## Clock Polarity :




positive edge triggered flops -> clock’s offstate = 0  => **clock polarity** +


negative edge triggered flop -> clock’s offsate = 1 => **clock polarity** +

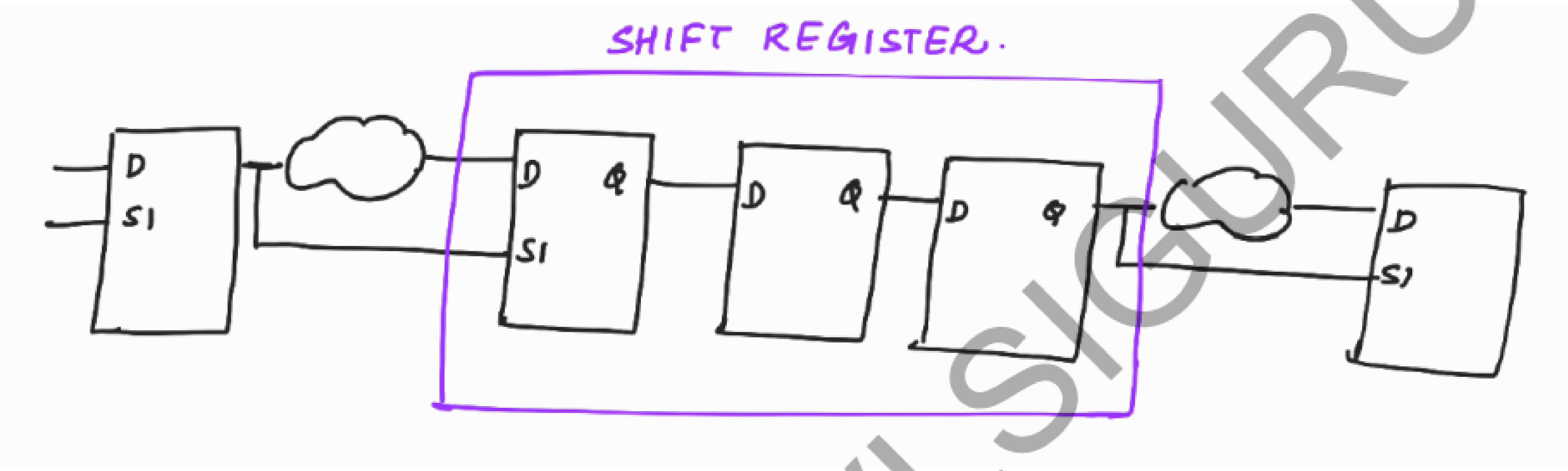
positive edge triggered flops -> clock’s offstate = 1  => **clock polarity** -

negative edge triggered flop -> clock’s offsate = 0 => **clock polarity** -

# Scan Cell Report (Contd…)



When there are shift registers in the design, the tool will replace only the first flop of the shift register as a scan flop (because it has to select between D and SI). The other flops in the shift register will be kept as normal flops.

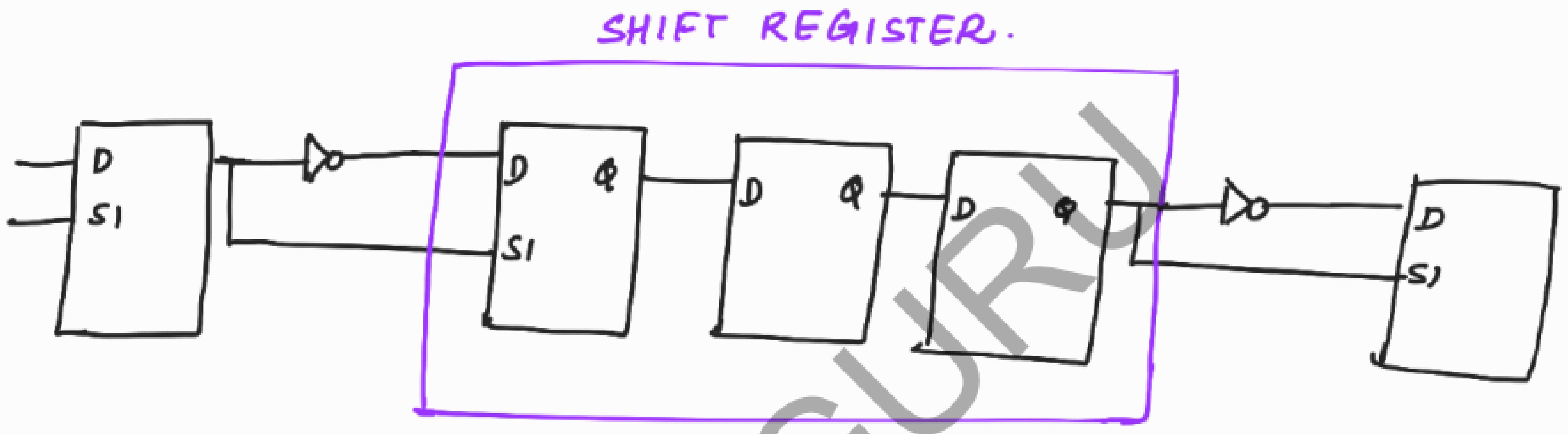



Shift register flops will also be a part of scan chain.

# Example :

no. of ffs in below scan chain = 5

Assume: 11011 (pattern to be loaded)

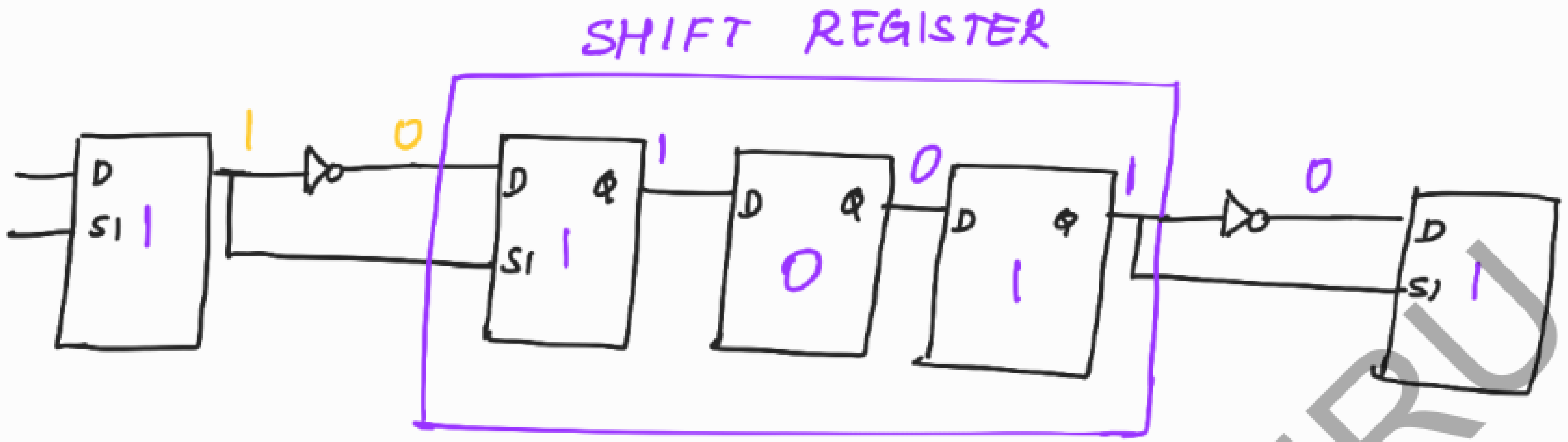



no. of clk pulses in shift = no. of ffs in the scan chain

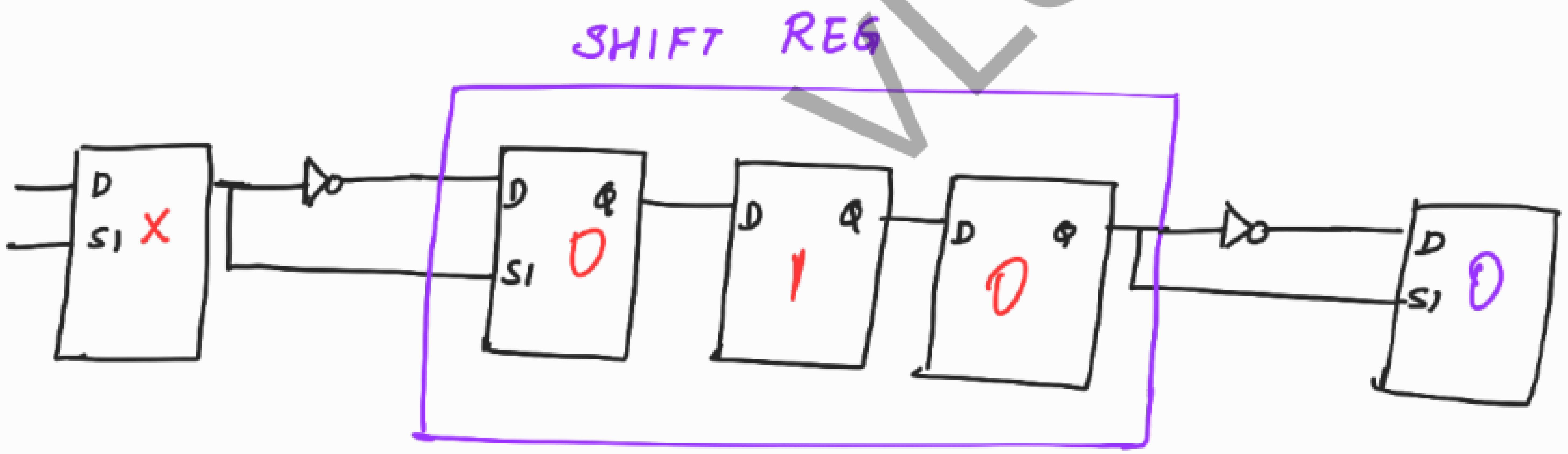
= 5

(After 5 pulses, the required pattern will be loaded into the scan chain)

## SHIFT




## CAPTURE



# OUTPUT FILES OF SCAN INSERTION

- Scan Inserted Netlist
- ATPG dofile
- ATPG testproc
- Scandef

# SCAN INSERTED NETLIST

- Design modified with inserted test structures (normal flops are replaced by their scan equivalents and scan chains are stitched).
- New ports added to the design : scan in, scan out, scan enable

# ATPG dofile

- Clock information
- Reset information
- Pin constraints

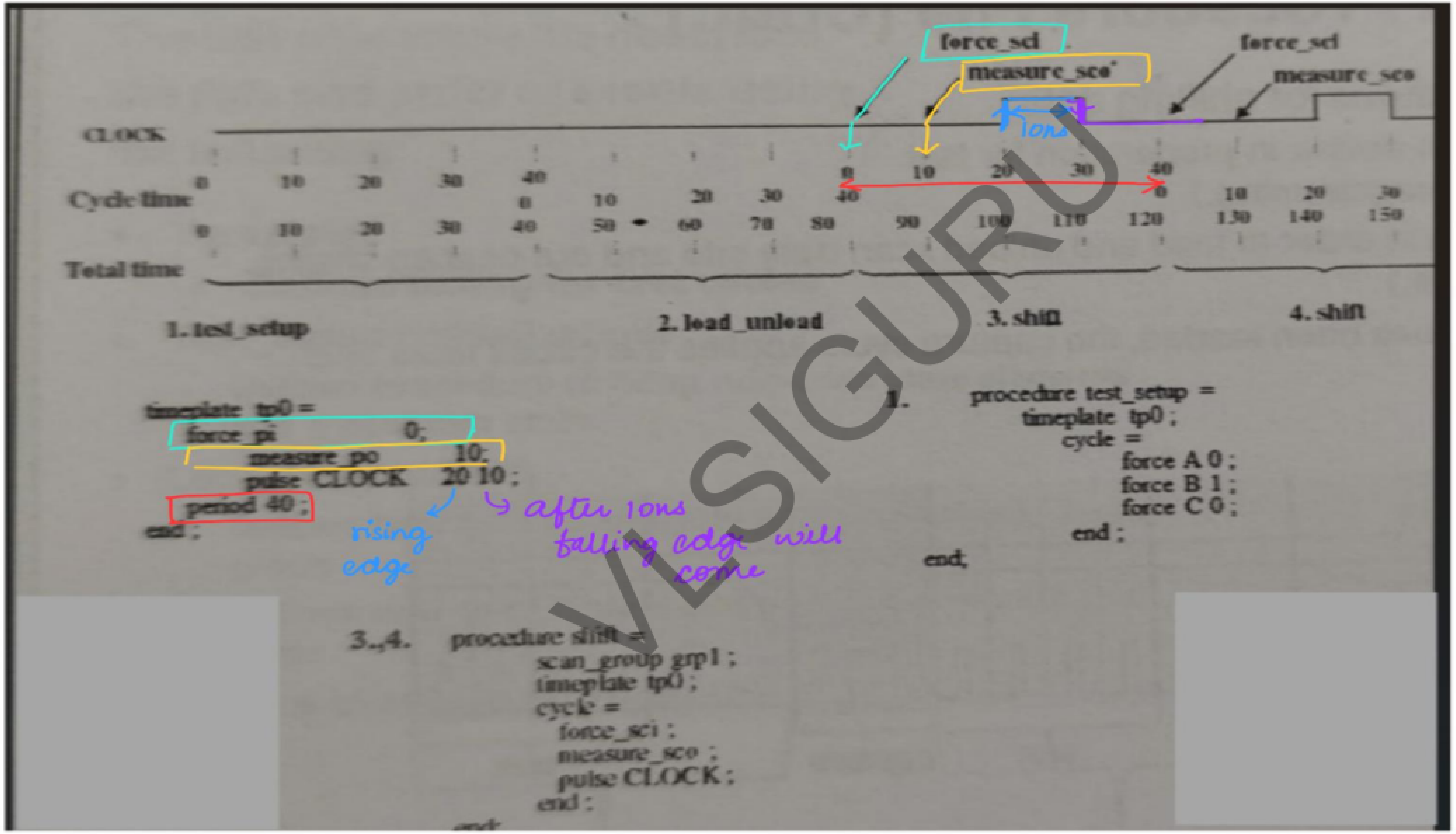
Line 7,8,9 → information about scan chains (chain no, scan in port, scan out port)

Line 6 → to apply the procedures given in the testproc file for all scan chains in the scan group.

Scan group – It is a set of scan chains that operate in parallel and share a common test procedure file.
The scan chains operating in parallel should have different scan inputs.

# ATPG TESTPROC FILE

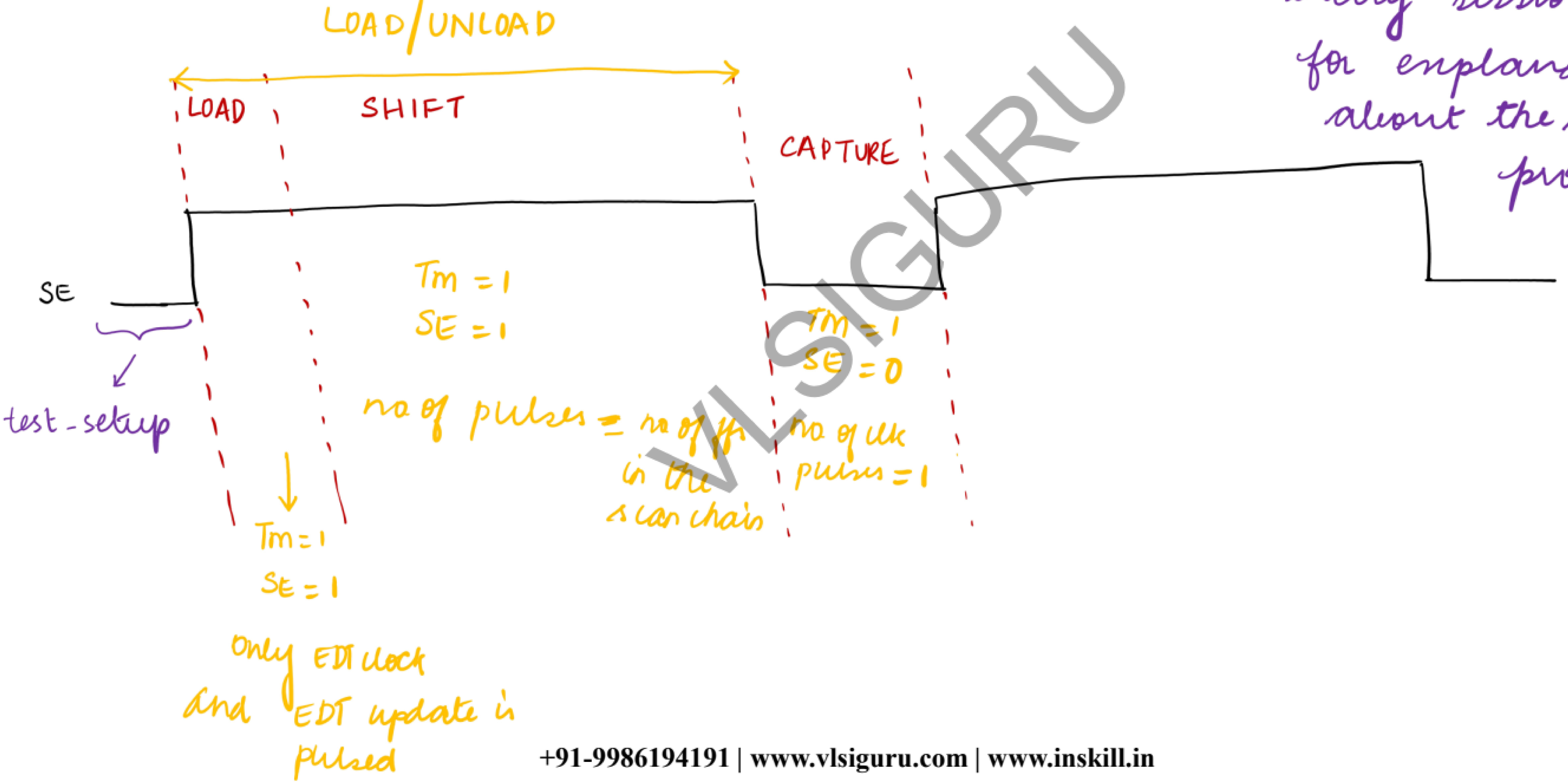
## TIMEPLATE EXPLANATION



# PROCEDURES IN TESTPROC FILE

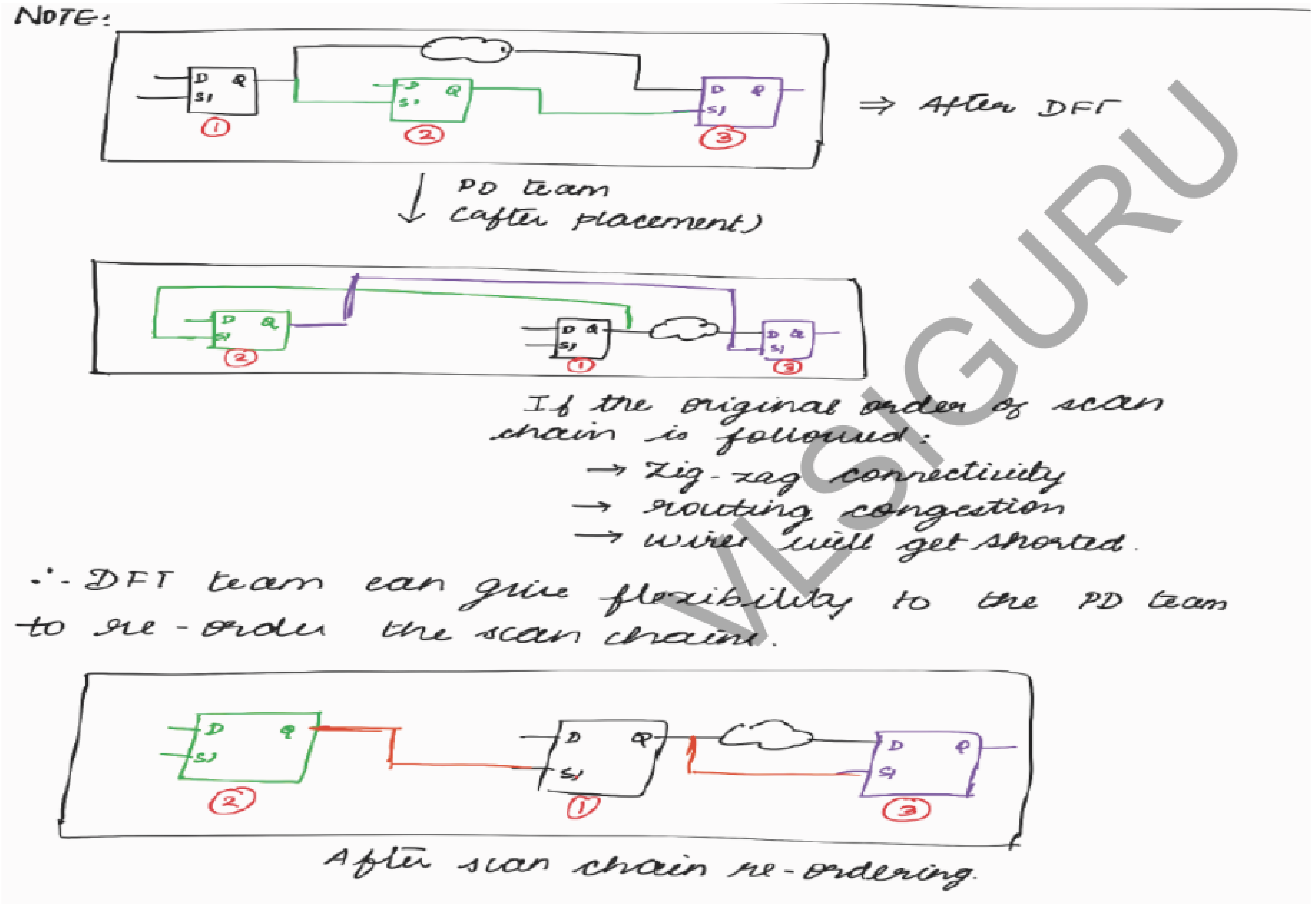


Please refer to the theory session notes for explanation about the various procedures.



# Scandef file

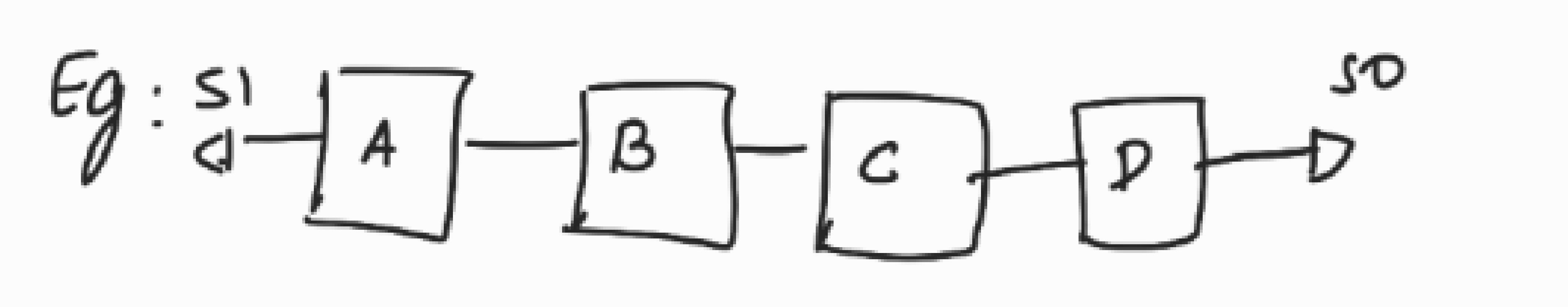
Scandef file is used by PD team during scan chain re-ordering.



# Scandef file (Contd...)



## Contents of Scandef file :




chain 1

Starting : A
↓
B
↓
C

Ending : D

Note:- ordered :- do not change order while re-ordering

floating :- can change order while re-ordering